WORKED SOLUTION

STEP I 2004, Question 4

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q4

The key idea

Whenever you see $\sqrt{x^2-1}$, think of the identity $\sec^2 t - 1 = \tan^2 t$: putting $x = \sec t$ turns the square root into $\tan t$, and since $\mathrm{d}x = \sec t \tan t \, \mathrm{d}t$, the $\tan t$ cancels too. Parts (ii) and (iii) look different, but completing the square inside the root turns each into the same shape, $\sqrt{u^2 - a^2}$ with $u = x + 2$, and the factor $x + 2$ outside the root is then exactly what the substitution needs.

Solution

The derivative

Writing $\sec t = (\cos t)^{-1}$ and using the chain rule,

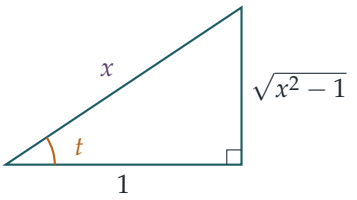
$$\frac{\mathrm{d}}{\mathrm{d}t} \sec t = -(\cos t)^{-2} \cdot (-\sin t) = \frac{\sin t}{\cos^2 t} = \sec t \tan t.$$

Part (i)

Put $x = \sec t$, so $\mathrm{d}x = \sec t \tan t \, \mathrm{d}t$. When $x = \sqrt{2}$, $\cos t = 1/\sqrt{2}$ and $t = \frac{\pi}{4}$; when $x = 2$, $\cos t = \frac{1}{2}$ and $t = \frac{\pi}{3}$. For t in $[\frac{\pi}{4}, \frac{\pi}{3}]$ we have $\tan t > 0$, so

$$\sqrt{x^2-1} = \sqrt{\sec^2 t - 1} = \sqrt{\tan^2 t} = \tan t$$

(with no modulus sign needed). The triangle below shows the same thing: hypotenuse x, adjacent side 1, so the opposite side is $\sqrt{x^2-1}$.



The integral becomes

$$\begin{aligned}\int_{\sqrt{2}}^{2} \frac{1}{x^3\sqrt{x^2-1}}\,\mathrm{d}x &= \int_{\pi/4}^{\pi/3} \frac{\sec t\tan t}{\sec^3 t\,\tan t}\,\mathrm{d}t = \int_{\pi/4}^{\pi/3} \cos^2 t\,\mathrm{d}t \\ &= \int_{\pi/4}^{\pi/3} \tfrac{1}{2}(1+\cos 2t)\,\mathrm{d}t = \left[\tfrac{1}{2}t + \tfrac{1}{4}\sin 2t\right]_{\pi/4}^{\pi/3} \\ &= \left(\frac{\pi}{6}+\frac{\sqrt{3}}{8}\right) - \left(\frac{\pi}{8}+\frac{1}{4}\right) = \frac{\sqrt{3}-2}{8} + \frac{\pi}{24},\end{aligned}$$

using $\cos^2 t = \frac{1}{2}(1+\cos 2t)$, $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$ and $\sin\frac{\pi}{2} = 1$. This is the required result.

Part (ii)

Ask yourself Part (ii) has no $\sqrt{x^2-1}$ in it. How can part (i) help?

Multiply out under the root and complete the square:

$$(x+1)(x+3) = x^2+4x+3 = (x+2)^2 - 1.$$

So with $u = x+2$ (and $\mathrm{d}u = \mathrm{d}x$) the integral is

$$\int \frac{1}{u\sqrt{u^2-1}}\,\mathrm{d}u.$$

The integrand is only defined when $u^2 > 1$. Take first $u > 1$ (that is, $x > -1$), and put $u = \sec t$ with $0 < t < \frac{\pi}{2}$, so that $\tan t > 0$ and $\sqrt{u^2-1} = \tan t$. Then

$$\int \frac{\sec t\tan t}{\sec t\,\tan t}\,\mathrm{d}t = \int 1\,\mathrm{d}t = t + C,$$

and since $\sec t = u$ we have $t = \operatorname{arcsec} u$ (equivalently $\arccos(1/u)$). Hence, for $x > -1$,

$$\int \frac{1}{(x+2)\sqrt{(x+1)(x+3)}}\,\mathrm{d}x = \operatorname{arcsec}(x+2) + C.$$

For completeness, the integrand is also defined for $x < -3$, where $u < -1$. There, put $u = -v$ with $v > 1$: the integrand $\dfrac{1}{u\sqrt{u^2-1}}\,\mathrm{d}u$ becomes $\dfrac{1}{(-v)\sqrt{v^2-1}}(-\mathrm{d}v) = \dfrac{1}{v\sqrt{v^2-1}}\,\mathrm{d}v$, which integrates to $\operatorname{arcsec} v$. Both cases are covered by

$$\operatorname{arcsec}|x+2| + C = \arccos\frac{1}{|x+2|} + C,$$

with possibly different constants on the two intervals.

Part (iii)

Again complete the square:

$$x^2+4x-5 = (x+2)^2 - 9,$$

so with $u = x + 2$ we need $\displaystyle\int \frac{1}{u\sqrt{u^2-9}}\,\mathrm{d}u$. This time the natural substitution is $u = 3\sec t$, chosen so that $u^2 - 9 = 9\tan^2 t$. For $u > 3$ (that is, $x > 1$) take $0 < t < \frac{\pi}{2}$, so $\sqrt{u^2-9} = 3\tan t$ and $\mathrm{d}u = 3\sec t \tan t\,\mathrm{d}t$. Then

$$\int \frac{3\sec t\tan t}{3\sec t \cdot 3\tan t}\,\mathrm{d}t = \int \frac{1}{3}\,\mathrm{d}t = \frac{t}{3} + C,$$

and $\sec t = u/3$. Hence, for $x > 1$,

$$\int \frac{1}{(x+2)\sqrt{x^2+4x-5}}\,\mathrm{d}x = \frac{1}{3}\operatorname{arcsec}\frac{x+2}{3} + C.$$

Exactly as in part (ii), the case $x < -5$ (where $u < -3$) is handled by $u = -v$, and both intervals are covered by $\frac{1}{3}\operatorname{arcsec}(\frac{1}{3}|x+2|) + C$.

Checks

- Simpson's rule on the integral in part (i) gives $0.0974060\ldots$, matching $\frac{1}{8}(\sqrt{3}-2) + \frac{\pi}{24}$.
- Differentiate the answers: $\displaystyle\frac{\mathrm{d}}{\mathrm{d}u}\operatorname{arcsec} u = \frac{1}{u\sqrt{u^2-1}}$ for $u > 1$ (put $u = \sec t$ and use the derivative from the start). For example, $\int_0^3$ of the part (ii) integrand is numerically 0.322241, and $\operatorname{arcsec} 5 - \operatorname{arcsec} 2 = \arccos\frac{1}{5} - \frac{\pi}{3}$ agrees.
- Part (iii) should be part (ii) rescaled: with $u = 3w$, $\int \frac{\mathrm{d}u}{u\sqrt{u^2-9}} = \frac{1}{3}\int \frac{\mathrm{d}w}{w\sqrt{w^2-1}}$, which explains the factor $\frac{1}{3}$.

Common slips

- Forgetting to change the limits in part (i), or getting them the wrong way round: $\sec t = \sqrt{2}$ gives $t = \frac{\pi}{4}$, not $\frac{\pi}{3}$.
- Writing $\sqrt{\tan^2 t} = \tan t$ without checking the sign of $\tan t$ on the range used.
- In part (iii), substituting $u = \sec t$ rather than $u = 3\sec t$, which leaves $\sqrt{\sec^2 t - 9}$ with no identity to simplify it.

Beyond STEP

The inverse secant is one member of a family: substitutions $x = a\sin t$, $a\tan t$, $a\sec t$ (or their hyperbolic cousins $a\sinh t$, $a\cosh t$) tame $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$ and $\sqrt{x^2-a^2}$, and completing the square reduces any $\sqrt{\text{quadratic}}$ to one of these. The care needed over signs and over which interval an antiderivative lives on is a first taste of how analysis courses make integration rigorous. Bartle and Sherbert, *Introduction to Real Analysis*, treats the substitution theorem properly, and Siklos, *Advanced Problems in Mathematics*, has more STEP integrals in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.